Introduction to AI: Team Final Project Guide

Course Project Requirements, Guidelines & Deliverables

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Project Overview

What This Project Is About

This semester-long team project challenges you to design, build, and evaluate an AI-powered system that addresses a real-world problem. You will apply concepts learned throughout the course — from classical machine learning to modern agentic AI — and communicate your findings through a formal IEEE-format technical report.

Key Learning Outcomes

- Identify a meaningful problem suitable for AI solutions
- Select and apply appropriate AI techniques
- Conduct rigorous experiments and analysis
- Communicate technical work in a professional format

Team Composition

2-3

Students Per Team

Strict Formation Rules — No Exceptions

✓ **Team Size: 2-3**
Optimal collaboration

✗ **No Solo Work**
Collaboration required

✗ **No Teams > 3**
Ensures equal work

Note: Each team member is expected to contribute meaningfully to all phases. Contribution statements must be included in the final report.

Mandatory AI Components

Component 1

Core AI Function (Choose ≥ 1)

- Machine Learning
- Computer Vision
- Search & Planning
- Natural Language Processing
- Optimization

+

Component 2

Agentic AI (Required)

An autonomous agent component that perceives its environment, makes decisions, and takes actions.

Examples:

- LLM-based agent
- Tool-using pipeline
- Multi-step reasoning system

What Is Agentic AI?

Systems That Perceive, Decide, and Act

Agentic AI refers to systems that operate with autonomy — taking sequences of actions, using tools, and pursuing goals without step-by-step human instruction.

Perception

Receives input from environment (text, images, sensors)

Reasoning

Applies logic to interpret situation & plan

Action

Executes response: calls tools, generates output

Memory

Retains context across multiple steps

Example Integrations:

- LLM agent retrieving & analyzing data
- Vision robot navigating obstacles
- Recommendation agent querying DBs

Project Workflow

Five Phases from Idea to Submission

Plan 01

Define problem, select AI methods, form team.

Project Proposal

Data 02

Collect, clean, and pre-process datasets.

Processed Dataset

Build 03

Implement AI functions and agentic component.

Working Codebase

Test 04

Run, diagnose, and compare algorithms.

Results & Figures

Report 05

Write IEEE-format technical report.

Final Submission

Each phase should be completed iteratively — revisit earlier phases as you learn more about your problem.

Phase 1: Planning

Start with a Clear Problem Definition

01 **What problem are you solving?**

Define the task precisely (e.g., "classify medical images as benign or malignant").

02 **Why does it matter?**

Motivate the real-world relevance and impact of the problem.

03 **Which AI function applies?**

Identify if it is classification, regression, search, NLP, or optimization.

04 **How does the agent fit?**

Describe what the agent perceives, decides, and executes autonomously.

05 **What data will you use?**

Identify a specific dataset (public repository or self-collected).

06 **How to measure success?**

Define evaluation metrics (accuracy, F1 score, BLEU, etc.).

A clear plan prevents wasted effort and keeps the team aligned.

Phase 2 — Data Pre-processing

Clean, Structured Data Is the Foundation of Every AI System

Feature Engineering

Numerical Features

Normalize to [0, 1] or standardize (zero mean, unit variance)

Textual Features

Bag-of-words, TF-IDF, or Embeddings (Word2Vec, BERT)

Image Features

Pixel intensity, Color histogram, HOG, or CNN embeddings

Data Splitting

Standard Split

Separate model fitting, tuning, and final evaluation

Cross-Validation

Use k-fold cross-validation when data is limited to ensure robustness

Data Augmentation

Image Data

Flipping, rotation, cropping, color jitter to increase diversity

Text Data

Synonym replacement, back-translation to improve generalization

Phase 3 — Implementation

Build a Reproducible Codebase

Language & Libraries

Python is recommended. You are encouraged to use established libraries (PyTorch, TensorFlow, scikit-learn) rather than implementing from scratch.

Self-Contained Submission

Code must run successfully when copied to a new folder. Do not rely on absolute paths or local environment configurations.

Documentation

Include a **README.md** with setup instructions. Use Git for version control to track collaboration.

AI Tool Policy

You are **welcome and encouraged** to use AI coding assistants (GitHub Copilot, ChatGPT, etc.) to accelerate development.

Must Disclose Usage in Report

Phase 4 — Experiments

Rigorous Experiments Distinguish Good Projects from Great Ones

Convergence Curves

Plot training loss/error vs. iterations to demonstrate model learning stability and speed.

Parameter Sensitivity

Analyze performance vs. key hyperparameters (e.g., learning rate, C, k) to justify choices.

Comparison Table

Quantitative metrics (Accuracy, F1, etc.) comparing your approach against baselines.

Qualitative Examples

Showcase specific cases where the system succeeds and fails to provide insight.

A complete project presents ALL FOUR result types and provides analysis explaining the patterns observed.

Phase 5 — Technical Report

Format Specs

Format Style

IEEE Conference Two-Column

Page Limit

Up to 8 pages (excluding references)

Requirement

No minimum page count — prioritize quality over quantity

Required Sections

01 Title & Authors

02 Abstract (100-300 words)

03 Introduction

04 Related Work

05 Methodology

06 Experiments & Results

07 Analysis & Discussion

08 Conclusion & Future Work

09 **AI Tool Disclosure**

10 References

AI Tool Disclosure

Using AI Tools Is Encouraged — Transparency Is Required

Which tools were used?

List specific tools (e.g., ChatGPT, GitHub Copilot, Grammarly, DALL·E).

For what purpose?

Explain the use case (e.g., code generation, debugging, writing assistance).

To what extent?

Describe the level of reliance (e.g., "drafted introduction, then substantially revised").

What was independent?

Clarify your own contribution (e.g., "experimental design and analysis were independent").

Undisclosed AI usage is considered a violation of academic integrity.

Recommended Topics

Broad Topic Areas Spanning All Five AI Functions

Machine Learning

- Predictive health diagnosis
- Fraud detection systems
- Personalized recommenders

Computer Vision

- Object detection apps
- Medical image analysis
- Autonomous navigation

NLP

- Sentiment analysis chatbot
- Document summarization agent
- Question Answering system

Search & Planning

- Puzzle-solving agent
- Route optimization
- Game-playing AI

Optimization

- Resource scheduling
- Supply chain optimization
- Hyperparameter tuning agent

Teams are encouraged to propose their own topics. Instructor approval is required for custom topics.

Evaluation Criteria

Projects Are Graded on Technical Depth, Rigor, and Communication

10% — **Problem Definition**
Clarity and significance of the chosen problem.

25% — **AI Implementation**
Correctness and sophistication of AI functions.

20% — **Agentic Component**
Quality and integration of the agentic AI element.

25% — **Experiments**
Rigor, completeness, and insight of experimental results.

15% — **Report Quality**
Adherence to IEEE format, clarity, and professionalism.

5% — **AI Disclosure**
Transparency and completeness of AI tool usage statement.

> *Note: All team members receive the same grade unless a peer evaluation indicates significantly unequal contribution.*

Important Resources

Datasets, Papers, and Tools to Get You Started

Datasets

Kaggle

Broad collection across all domains

Hugging Face

NLP and multimodal datasets

Papers With Code

Benchmarks with linked datasets

UCI ML Repository

Classical ML datasets

Agentic AI

LangChain

LLM-based agent pipelines

AutoGen

Multi-agent conversation framework

OpenAI Assistants

Tool-using LLM agents

Templates

IEEE Report Templates

Official LaTeX and Word templates will be provided directly on the course page.

Summary & Key Deadlines

Requirements Recap

- Team size: **2–3 students**
- Must include: **Core AI Function + Agentic AI**
- Report format: **IEEE two-column**, up to 8 pages
- AI tool usage: **Must be disclosed** in report

> *Questions? Reach out during office hours or post on the course forum.*

Suggested Milestones

Feb 2025

Team formation & topic selection
Project proposal (1 page)

March 2026

Data collection & pre-processing
Implementation

April 2026

Experiments & analysis

Final report submission